Predictive Analytics Models in Accounting & Finance

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Regression Model

- Relation between variables where changes in some variables may "explain" or possibly "cause" changes in other variables.
- Explanatory variables are termed the **independent** variables and the variables to be explained are termed the **dependent** variables.
- Regression model estimates the nature of the relationship between the independent and dependent variables.
  - Change in dependent variables that results from changes in independent variables, i.e., size of the relationship.
  - Strength of the relationship.
  - Statistical significance of the relationship.

Regression Models: Examples

- Dependent variable is the return on an asset – independent variable is the return on the market index (market model).
- Dependent variable is the excess return (over risk-free rate) – independent variable is the market excess return or market risk premium (CAPM model).
- Dependent variable is retail price of gasoline in Nova Scotia – independent variable is the price of crude oil.
- Dependent variable is employment income – independent variables might be hours of work, education, occupation, sex, age, region, years of experience, unionization status, etc.

- Price of a product and quantity produced or sold:
  - Quantity sold affected by price. Dependent variable is quantity of product sold – independent variable is price.
  - Price affected by quantity offered for sale. Dependent variable is price – independent variable is quantity sold.

Bivariate and Multivariate models

Bivariate or simple regression model

(Education) $x \longrightarrow y$ (Income)

Multivariate or multiple regression model

(Education) x_1
(Sex) x_2
(Experience) x_3
(Age) x_4

$\longrightarrow y$ (Income)

Model with simultaneous relationship

Price of wheat $\longleftrightarrow$ Quantity of wheat produced

Bivariate or simple linear regression

- x is the independent variable
- y is the dependent variable
- The regression model is:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

- The model has two variables, the independent or explanatory variable, x, and the dependent variable y, the variable whose variation is to be explained.
- The relationship between x and y is a linear or straight line relationship.
- Two parameters to estimate – the slope of the line β_1 and the y-intercept β_0 (where the line crosses the vertical axis).
- ε is the unexplained, random, or error component. Much more on this later.

Estimating the Regression Line

- The regression model is $y = \beta_0 + \beta_1 x + \varepsilon$
- Data about x and y are obtained from a sample.
- From the sample of values of x and y, estimates b_0 of β_0 and b_1 of β_1 are obtained using the least squares or another method.
- The resulting estimate of the model is: $\hat{y} = b_0 + b_1 x$
- The symbol $\hat{y}$ is termed "y hat" and refers to the predicted values of the dependent variable y that are associated with values of x, given the linear model. Notes:
  - Regression analysis does not establish a cause-and-effect relationship, just that there is a relationship.
  - The cause-and-effect relationship must be determined using a theoretical model or a logical reason.
  - The dependent variable is a continuous variable.
  - The independent variables can take any form - continuous or discrete or indicator variables.
  - The simple linear regression model has one independent variable.
  - The multiple linear regression model has two or more independent variables.

Simple Linear Model: OLS

- Please see the example on estimating the Market Model in Finance.

Outcome Variable and Predictive Model

Example 1: Predicting exam score

Outcome Variable: Exam score

- This is the variable we're interested in predicting or explaining. Exam scores can vary continuously and can take any value within a range (e.g., from 0 to 100).

Predictor Variable: Hours of study

- This is the variable we believe might influence or predict the outcome variable (exam scores). Hours of study can also vary continuously, such as 1 hour, 2.5 hours, 5 hours, etc.

Example 2: Predicting customer purchase decision

Outcome Variable: Purchase decision

- This variable is binary, with a value of 1 indicating that the customer made a purchase and a value of 0 indicating that the customer did not make a purchase.

Predictor Variables: Various customer characteristics.

- These are the variables we believe might influence or predict the purchase decision. They could include factors such as age, income, gender, etc.

Example 3: Predicting customer satisfaction level

Outcome Variable: Customer satisfaction level

- This is the variable we're interested in predicting or explaining. It represents the level of satisfaction a customer has with a product. This variable is ordered, with three levels: "Low," "Medium," and "High," where "Low" is considered the least satisfied level and "High" is considered the most satisfied level.

Predictor Variables: Various product features or characteristics

- These are the variables we believe might influence or predict customer satisfaction level. They could include factors such as product quality, price, customer service, etc.

Panel Data Models

- Panel data provide information on individual behavior, both across individuals and over time -- they have both cross-sectional and time-series dimensions.
- Panel data include N individuals observed at T regular time periods.
- Panel data can be balanced when all individuals are observed in all time periods ($T_i = T \; \forall \; i$) or unbalanced when individuals are not observed in all time periods ($T_i \neq T$) .
- We assume correlation (clustering) over time for a given individual, with independence over individuals.
- Example: the income for the same individual is correlated over time but it is independent across other individuals.

Panel Data Characteristics

- **Panel data types**
  - Short panel: many individuals and few time periods (we use this case in class)
  - Long panel: many time periods and few individuals
  - Both: many time periods and many individuals
- **Regressors:**
  - Varying regressors, X_{it}. For example, annual income for a person, annual consumption of a product.
  - Time-invariant regressors, $X_{it} = X_i \ \forall \ t$. For example, gender, race, education
  - Individual-invariant regressors, $X_{it} = X_t \ \forall \ i$. For example, time trend, economy trends such as unemployment rate

Panel Data Characteristics

- Individual mean: $\bar{x}_i = \frac{1}{T}\sum_t x_{it}$
- Overall mean: $\bar{x} = \frac{1}{NT}\sum_i \sum_t x_{it}$
- Overall variance: $S_0^2 = \frac{1}{NT-1}\sum_i \sum_t (x_{it} - \bar{x})^2$
- Between variance: $S_B^2 = \frac{1}{N-1}\sum_i (\bar{x}_i - \bar{x})^2$
- Within variance: $S_w^2 = \frac{1}{NT-1}\sum_i \sum_t (x_{it} - \bar{x}_i)^2 = \frac{1}{NT-1}\sum_i \sum_t (x_{it} - \bar{x}_i + \bar{x})^2$
- The overall variation can be decomposed into between variation and within variation: $S_o^2 = S_B^2 + S_w^2$

- Time-invariant regressors (race, gender, education) have zero within variation.
- Individual-invariant regressors (time, economy trends) have zero between variation.
- We need to check the data to see if the between or within variation is larger for each variable. You can use the xtsum command in STATA for that.

Panel Data Models

- **Pooled Model:**
  - The pooled model specifies constant coefficients, the usual assumptions for cross-sectional analysis.
  - The model is: $Y_{it} = \alpha + X_{it}'\beta + \mu_{it}$
  - This is the most restrictive panel data model and is not used much in the literature.
- **Individual-specific Effects Models:**
  - We assume that there is unobserved heterogeneity across individuals captured by α_i. For example, unobserved characteristics of an individual that affects his/her credit score.
  - The main question is whether the individual-specific effects α_i are correlated with the regressors. If they are correlated, we have the fixed effects model. If they are not correlated, we have the random effects model.

Panel Data Models: FE

- The FE model allows the individual-specific effects α_i to be correlated with the regressors X.
- We include α_i as intercepts in our regresison model.
- Each individual has a different intercept term and the same slope parameters.
- The FE model is: $Y_{it} = \alpha_i + X_{it}'\beta + \mu_{it}$
- We can recover the individual specific effects after estimation as: $\widehat{\alpha_i} = \bar{Y}_i - \bar{X}_i'\hat{\beta}$
- In other words, the individual-specific effects are the leftover variation in the dependent variable that cannot be explained by the regressors. Time dummies can be included in the regressors X.

Panel Data Models: RE

- The RE model assumes that the individual-specific effects α_i are distributed independently of the regressors.
- We include α_i in the error term.
- Each individual has the same slope parameters and a composite error term $\varepsilon_{it} = \alpha_i + e_{it}$
- The RE model is: $Y_{it} = X'_{it}\beta + (\alpha_i + e_{it})$
- In the model, $Var(\varepsilon_{it}) = \sigma_\alpha^2 + \sigma_e^2$, and $Cov(\varepsilon_{it}, \varepsilon_{is}) = \sigma_\alpha^2$. That means that the correlation structure is: $\rho_\varepsilon = Corr(\varepsilon_{it}, \varepsilon_{is}) = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma_e^2}$
- Rho is the inter-class correlation of the error. Rho is the fraction of the variance in the error due to the individual-specific effects. It approaches 1 if the individual effects dominate the idiosyncratic error.

Panel Data Models: OLS vs FE vs RE

- Breusch-Pagan Lagrange Multiplier test:
  - If the LM test is significant, use the random effects model instead of the OLS model. We still need to test for fixed versus random effects.

- Hausman test:
  - The random effects estimator is more efficient so we need to use it if the Hausman test supports it. If it does not support it, use the fixed effects model.
  - Hausman test tests whether there is a significant difference between the fixed and random effects estimators.

Panel Data Models: Example

- Please see the effect of your education and experience on your future wages example

Binary Outcome Models

- Binary outcome models are among the most used in applied economics and finance. For example, questions such as should the firm file for bankruptcy, should the firm pay any dividend, should the hire a new CFO – involve binary responses.

- The dependent variable is: $Y = \begin{cases} 0 \; if \; NO \\ 1 \; if \; Yes \end{cases}$

- Binary outcome models estimate the probability that $Y = 1$ as a function of the independent variables as follows:
- $p = prob[\![Y = 1 | \, X]\!] = F(X'\beta)$
- There are three different models depending on the functional form of $F(X'\beta)$.

Binary Outcome Models

- **Linear Probability Model:**
  - In the linear probability model, $F(X'\beta) = X'\beta$ and $p = prob[\![Y = 1|\, X]\!] = X'\beta$
  - A problem with the regression model is that the predicted probabilities will not be limited between 0 and 1.
  - We do not use the regression model with binary outcome data.
- **Logit Model:**
  - For the logit model, $F(X'\beta)$ the cdf of the logistic distribution and is given as follows:
  - $F = (X'\beta) = \Lambda(X'\beta) = \frac{exp^{X'\beta}}{1+exp^{X'\beta}}$
  - The predicted probabilities are limited between 0 and 1.
- **Probit Model:**
  - For the probit model, $F(X'\beta)$ the cdf of the standard normal distribution and is given as follows:
  - $F(X'\beta) = \phi(X'\beta) = \int_{-\infty}^{X'\beta} \phi(Z)dZ$
  - The predicted probabilities are limited between 0 and 1.

Logit/Probit Model Estimation

- Probit and logit models are estimated using the maximum likelihood method.
- Interpretation of coefficients :
  - An increase in X increases/decreases the likelihood that $Y=1$ (makes that outcome more/less likely). In other words, an increase in X makes the outcome of 1 more or less likely.
  - We interpret the sign of the coefficient but not the magnitude. The magnitude cannot be interpreted using the coefficient because different models have different scales of coefficients.
- When estimating probit and logit models, it is common to report the marginal effects after reporting the coefficients.
- The marginal effects reflect the change in the probability of $Y=1$ given a 1 unit change in an independent variable X.
- For the logit and probit models, the marginal effects are calculated as: $\frac{\partial p}{\partial X_j} = F'(X'\beta) \times \beta_j$

Logit/Probit Model Estimation

- Please see the example on estimating the Insurance Model in Finance.

Bivariate Probit/Logit Model

- Bivariate outcome example:
  - Individual decision whether to buy a car or not and whether to lease or self-finance the purchase.
  - Individual decision whether to work or not and whether to have children or not.
- The bivariate probit model is a joint model for two binary outcomes.
- These outcomes may be correlated, with correlation ρ.
- If the correlation turns out insignificant, then we can estimate two separate probit models, otherwise we have to use a bivariate probit model.
- The unobserved latent variables are presented as:
  - $Y_1^* = X_1'\beta_1 + e_1$
  - $Y_2^* = X_2'\beta_2 + e_2$
- The bivariate probit model specifies the outcomes as:
  - $Y_1 = \begin{cases} 1 \; if \; Y_1^* > 0 \\ 0 \; if \; Y_1^* \leq 0 \end{cases}$
  - $Y_2 = \begin{cases} 1 \; if \; Y_2^* > 0 \\ 0 \; if \; Y_2^* \leq 0 \end{cases}$
- Marginal effects and predicted values can be estimated similarly to those for the binary probit models.

Multinomial Logit/Probit Model

- Example: You can choose three potential career paths after your graduation from the MBAN, i.e., get a Data Analyst Job, pursue further studies such as a PhD in Finance & Accountng, or stay unemployed and go on a vacation in Europe.
- Multinomial outcome dependent variable:
  - The dependent variable Y is a categorical, unordered variable.
  - An individual may select only one alternative.
  - The choices/categories are called alternatives and are coded as j =1, 2, ..., m.
  - The numbers are only codes and their magnitude cannot be interpreted (use frequency for each category instead of means to summarize the dependent variable).
  - Advanced-level model: Conditional Logit and Mixed Logit

Ordered Probit/Logit Model

- Examples of ordered dependent variables:
  - Rating systems (poor, fair, good, excellent)
  - Employment (unemployed, part time, full time)
  - Debt Instrument Ranking (senior, junior, sophomore, freshman)
  - Exam Grades (A, B, C, D, F)
  - Bond ratings (AAA, AA, A, BBB, BB, B, CCC, CC, D)

- The categories for the dependent variables are rankings so the numbers do not make sense (even if they are coded as 0, 1, 2, 3, 4, the difference between the first and second outcome may not be the same as between the second and third).

Count Data Models

- The dependent variable is counts (a non-negative integer): Y = 0, 1, 2, 3, 4
- The sample is concentrated on a few small discrete values.
- We study the factors affecting the average number of the dependent variable.
- Examples:
  - Consumer demand: the number of products that a consumer buys on Amazon
  - Recreational data: the number of trips taken per year
  - Family economics: the number of children a couple has
  - Health demand: the number of doctors visits

Poisson Model

- The Poisson model predicts the number of occurrences of an event.
- The Poisson model states that the probability that the dependent variable Y will be equal to a certain number Y is:
  - $p(Y = y) = \frac{e^{-\mu}\mu^{y}}{y!}$
- For the Poisson model, μ is the intensity or rate parameter, where
  - $\mu = \exp(X_i'\beta)$
- Interpretation of the coefficients: one unit increase in X will increase/decrease the average number of the dependent variable by the coefficient expressed as a percentage.

Poisson Model

- Equi-dispersion property of the Poisson distribution: the equality of mean and variance:
  - $E(Y|X) = var(Y|X) = \mu$
- This is a restrictive property and often fails to hold in practice, i.e., there is "over-dispersion" in the data. In this case, use the negative binomial model.
- Excess zeros problem of the Poisson distribution: there are usually more zeros in the data than a Poisson model predicts. In this case, use the zero-inflated Poisson model.

Negative Binomial Model

- The negative binomial model is used with count data instead of the Poisson model if there is over-dispersion in the data.
- Unlike the Poisson model, the negative binomial model has a less restrictive property that the variance is not equal to the mean (μ). More generally:
  - $var(Y|X) = \mu + \alpha\mu^2$
- Another functional form is $VAR(y|x) = \mu + \alpha\mu$, but this form is less used.
- The negative binomial model also estimates the over-dispersion parameter μ.
- Test for overdispersion
  - Estimate the negative binomial model which includes the over-dispersion parameter α and test if †is significantly different than zero: $H_0: \alpha = 0$ & $H_a: \alpha \neq 0$
- We have three cases:
  - When $\alpha = 0$, the Poisson model.
  - When $\alpha > 0$, over-dispersion (frequently holds with real data).
  - When $\alpha < 0$, under-dispersion (not very common).

Zero-inflated Poisson Model

- The zero-inflated model is used with count data when there is an excess zeros problem.
- The zero-inflated model lets the zeros occur in two different ways: as a realization of the binary process (z=0) and as a realization of the count process when the binary variable z=1.
- Example: you either like hiking or you do not. If you like hiking, the number of hiking trips you can take is 0, 1, 2, 3, etc. So you may like hiking, but may not take a trip this year. We are able to generate more zeros in the data.
- If the process generating the zeros is $f_1(.)$and the process generating the positive responses is $f_2(.)$ then the zero-inflated model is:
- $$g(Y) = \begin{cases} f_1(0) + \big(1 - f_1(0)\big) f_1(0) \; if \; Y = 0 \\ \big(1 - f_1(0)\big) f_2(Y) \;\; if \; Y \geq 1 \end{cases}$$
- The zero-inflated model is less frequently used than the hurdle model.
- The zero-inflated models can handle the excess zeros problem.

Other Model

- Principal Component Analysis (PCA)
- Propensity Score Matching
- Quantile Regression
- Survival Analysis Models
- Time Series Models
- Instrumental Variables Analysis

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